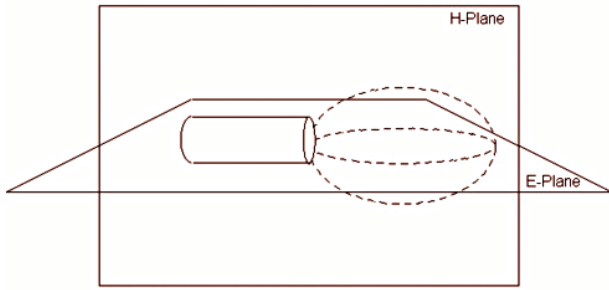
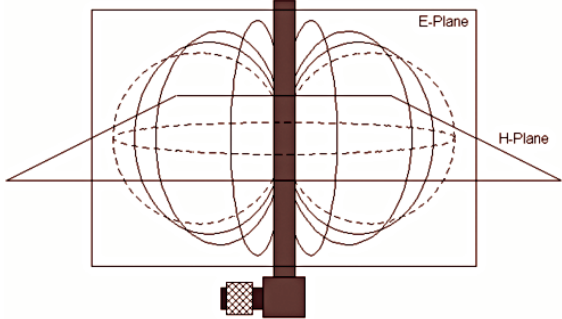


TABLE 2
RELATION BETWEEN AZIMUTH/ELEVATION AND E/H PLANE

Electric field or E plane	Magnetic field or H plane
Plane containing the electric field vector and the direction of maximum radiation	Plane containing the magnetic field and the direction of maximum radiation
Called as E aperture	Called as H aperture
Determines the polarisation or orientation of the radio wave	Lies at a right angle to the 'E' plane
For horizontal polarised antenna	
Coincides with the horizontal/Azimuth/XY plane	Coincides with the vertical/Elevation plane
For vertical polarised antenna	
Coincides with the vertical/YZ/Elevation plane	Coincides with the horizontal/Azimuth plane
Relation between E and H plane: (a) Horizontally polarised-directional antenna (Yagi-Uda), (b) vertically polarised-omni directional antenna (dipole)	
 	
(a)	(b)

radiation efficiency as a function of radial distance and angular position from the antenna. An antenna radiation pattern is the mathematical graphical representation of the radiation energy distribution into all directions of space and hence is three-dimensional, expressed in the spherical coordinate system.

Two principal planes

Characterising a well-behaved antenna's radiation properties with two plane patterns in a coordinate system involves azimuth plane pattern and elevation plane pattern. Assuming that the antenna is measured in the orientation in which it will be used, as shown in Fig. 1(a), the x-y plane or 'the horizontal' ($\theta = 90^\circ$) is the azimuth plane, and the pattern is measured by traversing the entire x-y plane. The plane orthogonal to the x-y plane, say the y-z plane or "the vertical" ($\phi = 90^\circ$), is the elevation plane in which the pattern is made by tra-

versing the entire y-z plane around the antenna under test. The elevation plane is oriented at right angles to the azimuth plane. In short, the principal plane patterns are:

- Azimuth/XY/horizontal plane (top - down view) (ϕ)
- Elevation/vertical/YZ plane (side - on view) (90°)

2D and 3D view

The radiation patterns can be three-dimensional (3D), i.e., a function of (θ, ϕ, r), or two-dimensional (2D). In 2D, the radiation pattern represents a cut of the 3D radiation pattern, for given angles $\theta = \theta_0$ or $\phi = \phi_0$. These two planar patterns (azimuth and elevation plane patterns) are used to describe radiation in a 3D pattern using spherical coordinates, as in Fig. 1(b), and are frequently shown as 2D plots using polar coordinates, as shown in Fig. 1(c).

Below is a summary comparing 3D and 2D patterns. A 2D polar plot can be preferred for symmetri-

cal directional antennas, but for omni-directional or newly designed evaluation, a 3D spherical pattern is used. The preference and comparison between 2D and 3D are shown in Table 1.

E and H plane

The orientation of the electrical and magnetic components of electromagnetic waves, called polarisation from a radiating antenna, is in a single linear direction for linearly polarised antennas. When representing the radiation patterns of linearly polarised antennas, the E and H planes can be used and compared with azimuth/elevation plane as in Table 2.

An antenna with a poor radiation pattern but high gain may perform less reliably and effectively than an antenna with an excellent radiation pattern but lower gain. Antenna gain can only act on the antenna's existing radiation pattern, which is again determined by the quality and details of the antenna's design. **EFY**